



Advanced Algorithms

COMP4121

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Database Access Problem

Database Access Problem

(More details can be found on pages 707-714 in *Algorithm Design* by Kleinberg and Tardos.)

- Assume that n processes want to access a database, and that the time is discrete, $t = 1, 2, 3, \dots$
- If at any instant t two processes (simultaneously) request access, there is a conflict and all processes are locked out of access.
- We assume that the processes cannot communicate with each other to agree on a joint strategy; each process has to decide if it will request access to database on its own, in isolation of other processes.
- One possible approach in such a situation is that each process at each instant t tosses a coin which produces outcome “request access” with a probability p and “do not request access” with probability $1 - p$.
- How should we choose p to maximise the probability of a successful access to the database for a process at any instant t ?

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- For a given probability p , the probability of the success of a process i at any instant t is $\mathcal{P}(\mathcal{S}(i, t)) = p(1 - p)^{n-1}$, because a process i requests access with probability p and the probability that no other process has requested access is $(1 - p)^{n-1}$.
- For what value of p does $\mathcal{P}(\mathcal{S}(i, t))$ have the largest possible value?
- Extremal points are found by solving

$$\frac{d}{dp}\mathcal{P}(\mathcal{S}(i, t)) = (1 - p)^{n-1} - p(n - 1)(1 - p)^{n-2} = 0$$

- Dividing both sides by $(1 - p)^{n-2}$ we get that the above equality holds just in case $1 - p - p(n - 1) = 0$ which is equivalent to $p = 1/n$, as one might expect.

- So for the optimal value $p = 1/n$ the probability of success for a process i at an instant t is

$$\mathcal{P}(\mathcal{S}(i, t)) = p(1 - p)^{n-1} = \frac{1}{n} \left(1 - \frac{1}{n}\right)^{n-1}$$

- To establish the asymptotic behaviour of $\mathcal{P}(\mathcal{S}(i, t))$ the following two facts are useful; both can be proved by establishing the signs of the relevant first derivatives:
 - 1 $\left(1 - \frac{1}{n}\right)^n$ increases monotonically from $1/4$ up to $1/e$ as n increases from 2 to ∞ ($e = 2.718\dots$).
 - 2 $\left(1 - \frac{1}{n}\right)^{n-1}$ decreases monotonically from $1/2$ down to $1/e$ as n increases from 2 to ∞ .

- Thus, since we had

$$\mathcal{P}(\mathcal{S}(i, t)) = \frac{1}{n} \left(1 - \frac{1}{n}\right)^{n-1}$$

we obtain

$$\frac{1}{n} \cdot \frac{1}{e} \leq \mathcal{P}(\mathcal{S}(i, t)) \leq \frac{1}{n} \cdot \frac{1}{2}$$

- Thus $\mathcal{P}(\mathcal{S}(i, t)) = \Theta\left(\frac{1}{n}\right)$.
- So after t many instants, what is the probability that a particular process has succeeded?
- It is actually easier to estimate the probability of failure after t instants:

$$\mathcal{P}(\text{failure after } t \text{ instants}) = \left(1 - \frac{1}{n} \left(1 - \frac{1}{n}\right)^{n-1}\right)^t$$

- Using the second of our two inequalities we get

$$\mathcal{P}(\text{failure after } t \text{ instants}) \approx \left(1 - \frac{1}{en}\right)^t$$

- We can now observe a somewhat strange phenomenon:
 - If we chose $t = en$ many consecutive instants (we ignore floors or ceilings necessary to make quantities integers) then the probability of a failure is quite large, because

$$\mathcal{P}(\text{failure after } t = en \text{ instants}) \approx \left(1 - \frac{1}{en}\right)^{en} \approx \frac{1}{e}$$

which is quite large, larger than $1/3$.

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- However, if we increase the number of instants only slightly, by taking $t = e n \cdot 2 \ln n$, then

$$\begin{aligned}\mathcal{P}(\text{failure after } t = e n \cdot 2 \ln n \text{ instants}) &\approx \left(1 - \frac{1}{e n}\right)^{e n \cdot 2 \ln n} \\ &= \left(\left(1 - \frac{1}{e n}\right)^{e n}\right)^{\ln n^2} \\ &\approx \left(\frac{1}{e}\right)^{\ln n^2} \\ &= \frac{1}{n^2}\end{aligned}$$

- Thus, a slight increase in the number of time instants, from $e n$ to $2 e n \ln n$ caused a dramatic reduction in the probability of failure of a process to access the database.

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- What is the probability that at least one process has failed to access data base after $2e n \ln n$ instants?
- If probability of failure of each process is less than $1/n^2$ and there are n processes, then the probability that at least one process failed cannot be larger than $n \cdot \frac{1}{n^2} = \frac{1}{n}$.
- Thus after $2e n \ln n$ instants all processes succeeded to access the data base with probability at least $1 - 1/n$.
- So if the processes could communicate and agree upon the order in which they would access the database, it would take n instants for all of them to access the database.
- If they use randomised process which does not require any communication, then they will be able to access the database with probability $1 - 1/n$ in time which is larger only by a relatively small factor of $2e \ln n$.